

Lab 3 –Subnetting (CH5)

Part 1: Intro to Subnetting

Goal: Calculate the number of subnets and hosts

- Given a **Class C** network and a number of subnets required, complete the table to identify the number of bits to borrow from the host field for the subnet field and the maximum number of host addresses available per subnet **(5 POINT)**.

Number of Subnets Required	Number of bits to borrow for the subnet field	Maximum number of hosts per subnet
2	1	$2^7-2=126$
5	3	$2^5-2=30$
12	4	$2^4-2=14$
24	5	$2^3-2=6$
40	6	$2^2-2=2$

- Given a **Class B** network and a number of subnets required, complete the table to identify the number of bits to borrow from the host field for the subnet field and the maximum number of host addresses available per subnet **(5 POINT)**.

Number of Subnets Required	Number of bits to borrow for the subnet field	Maximum number of hosts per subnet
5	3	$2^{13}-2=8,190$
8	3	$2^{13}-2=8,190$
35	6	$2^{10}-2=1,022$
200	8	$2^8-2=254$
400	9	$2^7-2=126$

- Given a **Class A** network and a number of subnets required, complete the table to identify the number of bits to borrow from the host field for the subnet field and the maximum number of host addresses available per subnet **(5 POINT)**.

Number of Subnets Required	Number of bits to borrow for the subnet field	Maximum number of hosts per subnet
10	4	$2^{20}-2=1,048,574$
20	5	$2^{19}-2=524,286$
80	7	$2^{17}-2=131,070$
400	9	$2^{15}-2=32,766$

2000	11	$2^{13}-2=8190$
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Goal: Given an IP address and subnet mask, calculate the maximum number of subnets that can be defined and the number of hosts on each subnet. Calculate the subnet that the IP address is on. Determine all of the subnets that can be defined for the network. For each subnet, find the range of host addresses and the broadcast address.

4. Given a **Class A** network and the prefix in the first column of the table, complete the table to identify the decimal and binary format subnet mask and the maximum number of host addresses available per subnet **(5 POINTS)**.

Prefix	Decimal Format Subnet Mask	Binary Format Subnet Mask	Maximum Number of Hosts per Subnet
/20	255.255.240.0	11111111.11111111.11110000.00000000	4094
/21	255.255.248.0	11111111.11111111.11111000.00000000	2046
/22	255.255.252.0	11111111.11111111.11111100.00000000	1022
...	...	...	...
/29	255.255.255.248	11111111.11111111.11111111.11111000	6
/30	255.255.255.252	11111111.11111111.11111111.11111100	2

5. Assume that you have been assigned the **172.25.0.0/16** network. You need to create 12 subnets. (7 points total)

#	Question	Answer
0	Which class is this? (1 point)	Class B
A	How many bits do you need to borrow from the host field for the subnet field? (1 point)	4 bits ($2^4=16>12$ subnets)
B	What is the increment for the subnet creation (1 points)	16
C	What is the maximum number of subnets that can be created with this number of bits? (1 point)	$2^4=16$
D	How many bits can be used to create the host space? (1 point)	12
E	What is the maximum number of host addresses available per subnet? (1 point)	$2^{12}-2=4094$
F	What prefix would you use? What is the subnet mask, in binary and decimal format? (1 point)	Prefix: /20 Decimal: 255.255.240.0 Binary: 11111111.11111111.11110000.00000000

- g. Complete the following table; define each of the subnets, the range of hosts addresses on the subnet and the directed broadcast address on the subnet (3 POINTS).

Subnet Number	Subnet Address	Range of Host Addresses	Direct Broadcast Address
0	172.25.0.0	172.25.0.1 - 172.25.15.254	172.25.15.255
1	172.25.16.0	172.25.16.1 - 172.25.31.254	172.25.31.255
2	172.25.32.0	172.25.32.1 - 172.25.47.254	172.25.47.255
3	172.25.48.0	172.25.48.1 - 172.25.63.254	172.25.63.255
4	172.25.64.0	172.25.64.1 - 172.25.79.254	172.25.79.255
5	172.25.80.0	172.25.80.1 - 172.25.95.254	172.25.95.255
6	172.25.96.0	172.25.96.1 - 172.25.111.254	172.25.111.255
7	172.25.112.0	172.25.112.1 - 172.25.127.254	172.25.127.255
8	172.25.128.0	172.25.128.1 - 172.25.143.254	172.25.143.255
9	172.25.144.0	172.25.144.1 - 172.25.159.254	172.25.159.255
10	172.25.160.0	172.25.160.1 - 172.25.175.254	172.25.175.255
11	172.25.176.0	172.25.176.1 - 172.25.191.254	172.25.191.255
12	172.25.192.0	172.25.192.1 - 172.25.207.254	172.25.207.255
13	172.25.208.0	172.25.208.1 - 172.25.223.254	172.25.223.255
14	172.25.224.0	172.25.224.1 - 172.25.239.254	172.25.239.255
15	172.25.240.0	172.25.240.1 - 172.25.255.254	172.25.255.255
...	...	...	...
Last subnet number	172.25.240.0	172.25.240.1 - 172.25.255.254	172.25.255.255

